

Lecture # 07

Amdahl's Law

PARALLEL AND DISTRIBUTED COMPUTING

CS-482



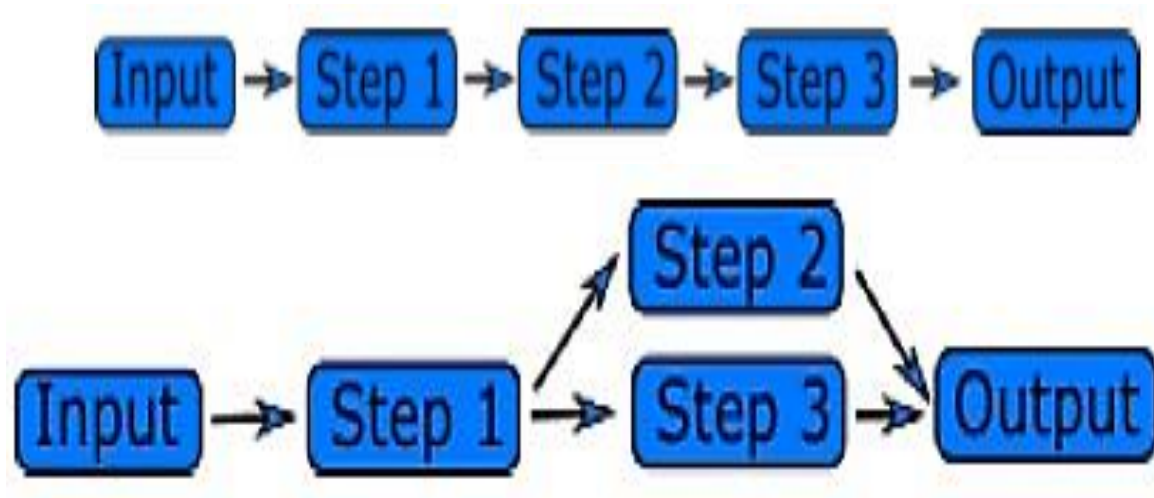
Amdahl's law

For a given fixed amount of computation W , the speedup of machine 2 relative to machine 1 is $S = \frac{T_1}{T_2}$

- T_1 is the time taken by machine 1 to perform W whereas T_2 is the time taken by machine 2 to perform the same amount of computation ($T_2 < T_1$).
- In 1967, Gene Amdahl pointed out, in a short paper,
 - the inherent limitations in trying to improve computer-system performance
 - by using multiple processors.
- The concept has come to be known as Amdahl's law.

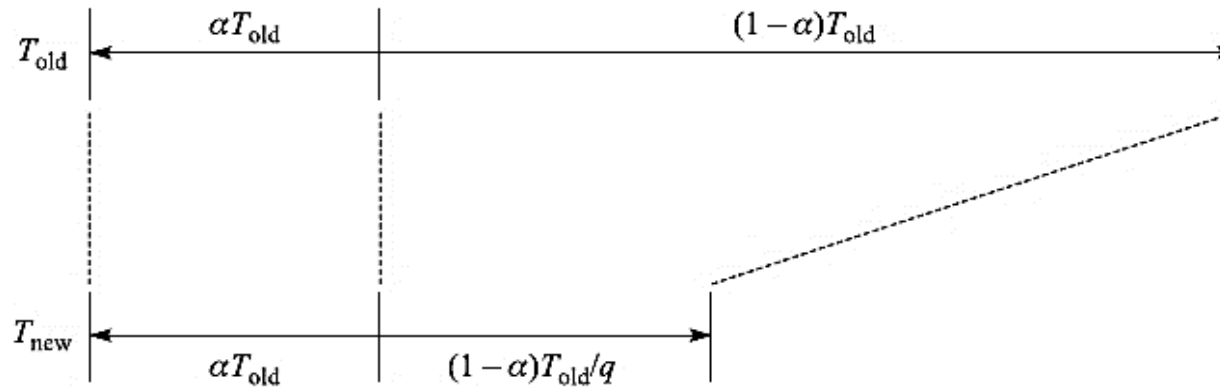
Amdahl's law

- Amdahl's argument is essentially that
 - the overall performance improvement observed in an application program
 - (with fixed amount of computation W)
 - is limited by that portion of the application
 - that is unaffected by whatever change was made to the system.



Amdahl's law

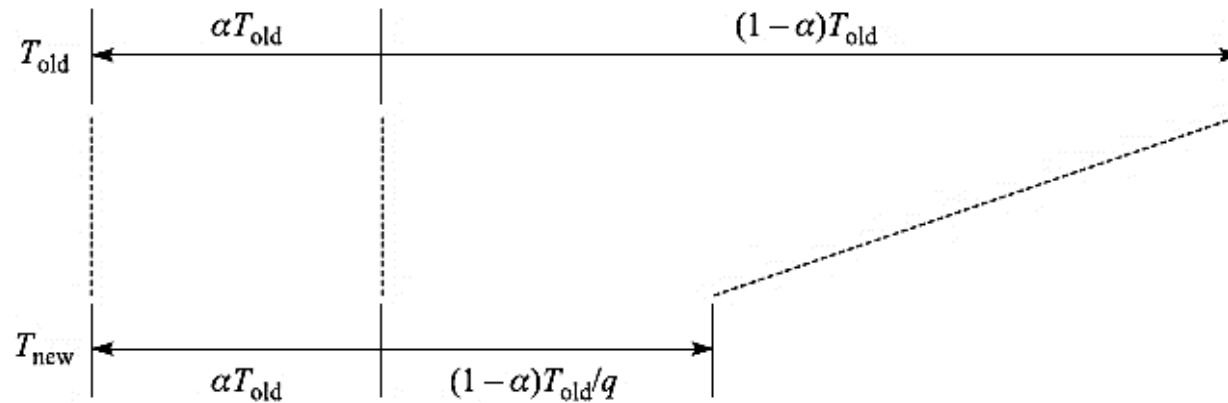
- Consider the execution time lines shown in fig 2.



- The top line shows the time, T_{old} , required to execute some program P on the system before any changes are made.
- Now assume that some change is made to the system that reduces execution time for some operations by a factor of q .
- The program now runs in time T_{new} , where $T_{new} < T_{old}$, as shown in the bottom line.

Amdahl's law

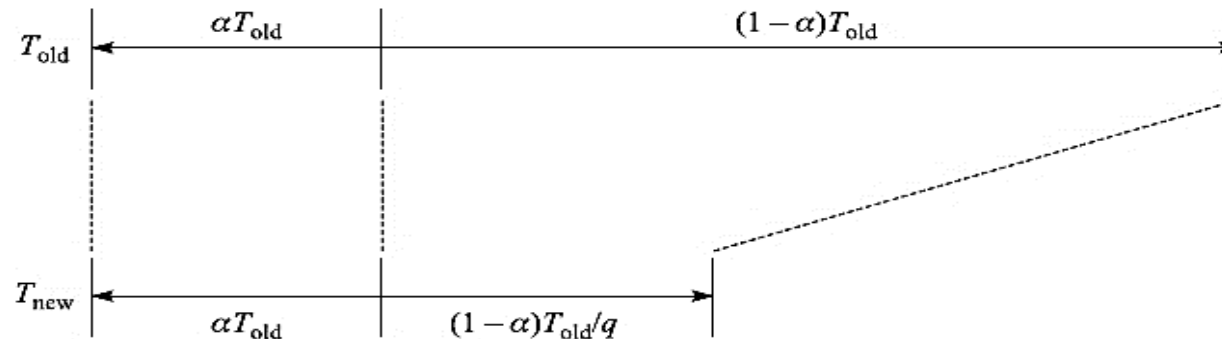
- Hence there are many other operations in the program that are unaffected by this change.



- Let α be the fraction of all operations that are unaffected by the enhancement.
- Then, as shown in the bottom line of fig 2, the new execution time, T_{new} , can be divided into two components.

Amdahl's law

- The first component, αT_{old} , is the execution time of that fraction of the program that is unaffected by the change.
- The second component of T_{new} , which is the remaining fraction $1-\alpha$ of the original execution time, has its performance improved by the factor q .
- Thus, the time required for this component is $(1-\alpha) T_{\text{old}}/q$.
- The overall speedup caused by this improvement is then found to be
- $$S = \frac{T_{\text{old}}}{T_{\text{new}}} = \frac{T_{\text{old}}}{\alpha T_{\text{old}} + (1-\alpha) T_{\text{old}}/q} = \frac{1}{1/q + \alpha(1-1/q)}$$



Amdahl's law

- This equation can be used to calculate the overall speedup obtained due to some improvement in the system, assuming that q and α can be determined.
- However, it is interesting to ask what happens as the impact on performance of the improvement becomes large, that is, as $q \rightarrow \infty$.
- It is easy to show that, in the limit as $q \rightarrow \infty$, $(1-\alpha) T_{old}/q \rightarrow 0$.
- Thus, the overall speedup, S , is bounded by $1/\alpha$.
- That is,

$$\lim_{q \rightarrow \infty} s = \lim_{q \rightarrow \infty} \frac{1}{1/q + \alpha(1 - 1/q)} = \frac{1}{\alpha}$$

Amdahl's law

- This result says that, no matter how much one type of operation in a system is improved,
- the overall performance is inherently limited by the operations that are unaffected by the improvement.
- For example, the best speedup that could be obtained in a parallel computing system with p processors is p .
- However, if 10% of a program cannot be executed in parallel, the overall speedup when using the parallel machine is at most $1/\alpha = 1/0.1=10$, even if an infinite number of processors were available.

Amdahl's law

- An obvious corollary to Amdahl's law
 - any system designer or programmer should concentrate on making the common case fast.
- That is, operations that occur most often will have the largest value of α .
- Thus, improving these operations will have the biggest impact on overall performance.
- Interestingly, the common cases also tend to be the simplest cases.
- As a result, optimizing these cases first tends to be easier than optimizing the more complex, but rarely used, cases.

Scaling Amdahl's law

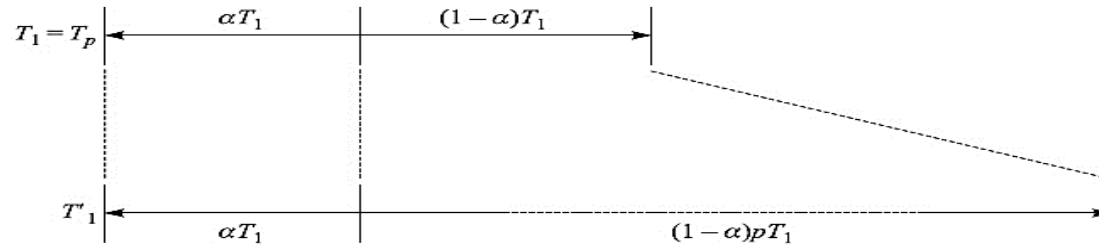
- One of the major criticisms concerning Amdahl's law has been that it emphasizes the wrong aspect of the performance potential of parallel-computing systems.
- The argument is that purchasers of parallel systems want to solve larger problems within the *available time*.
- Following this line of argument leads to the following “scaled” or “fixed-time” version of Amdahl's law.
- It is common to judge the performance of an application executing on a parallel system by
 - comparing the parallel execution time with p processors, T_p ,
 - with the time required to execute the equivalent sequential version of the application program, T_1 ,
 - using the speedup $S_p = T_1/T_p$.

Scaling Amdahl's law

- With the fixed-time interpretation, however, the assumption is that
 - there is no single-processor system that is capable of executing an equivalent sequential version of the parallel application.
 - The single-processor may not have a large enough memory, for example, or
 - the time required to execute the sequential version would be unreasonably long.

Scaling Amdahl's law

- In this case, the parallel-execution time is divided into
 - the parallel component, $1-\alpha$, and
 - the inherently sequential component, α , giving
 - $T_p = \alpha T_1 + (1-\alpha)T_1$ as shown in fig 3 below.



- T_1 is the time in which user wants to run application.
- Since no single-processor system exists that is capable of executing an equivalent problem of this size,
 - it is assumed that the parallel portion of the execution time would increase by a factor of p
 - if it were executed on a hypothetical single-processor system.

Scaling Amdahl's law

- This gives a hypothetical sequential execution time

$$T'_p = \alpha T_1 + (1-\alpha)pT_1.$$

- The parallel speedup is found to be

$$S_p = \frac{T'_1}{T_p} = \frac{\alpha T_1 + (1-\alpha)pT_1}{T_1} = p + \alpha(1-p).$$

- Thus, with this *fixed-time, variable-problem-size* form of Amdahl's law, the performance of a parallel machine is not limited by the portion that is inherently sequential.

Homework 1

An industrial process simulation involves five steps which are performed sequentially on system A. The steps 1 and 2 take 1 and 2 minutes respectively whereas steps 3 – 5 take 3 minutes each. Then system A was enhanced by introducing some parallelism to get a system B. The steps 3 to 5 on B now can be executed in parallel so that they take an overall time of 4 minutes whereas steps 1 and 2 are still to be executed in sequence. Calculate speedup using appropriate form of Amdahl's law.

Homework 2

An industrial process simulation is to be executed on a system in the available time, t_a , of 10 min which includes parallel execution time (on 10 processors) of 7.5 min. It has been estimated that the parallel portion of the execution time would increase by a factor of 12 if it were executed on a hypothetical single-processor system. Calculate the parallel speedup.

Homework 3

In designing a new computer system, we make an enhancement that improves some mode of execution by a factor of 10. This enhancement takes 50% of the time when the enhanced mode is in use. (Recall that Amdahl's law uses the fraction of the original, unenhanced execution time to find speedup)

- a) What is the speedup that we have obtained by using this fast mode?
- b) What percentage of the original execution time has been converted to fast mode?